

LAB EXPERIMENT RECORD

Set 21: Energy Consumption Analysis

Department	B.Tech AIDS
Course Code / Course Name	DSA06 / Data Handling and Visualization
Experiment	Data Visualization using R
Name	PRANAYAA S
Register Number	192524110
Date	13.10.2026
Day	Tuesday

Aim

To import and analyze energy consumption data in R using a histogram, density plot, bubble scatter plot, and bar chart, and to design an interactive dashboard with a monthly trend line chart and a sector-region heatmap using R Shiny.

Dataset: Energy_Consumption_Data.csv

Sector	Region	Month	Temp. (°C)	Units (kWh)	Cost (₹)	Renewable (%)	Peak Hours
Residential	North	Jan	15	320	2100	22	4
Commercial	South	Jan	24	540	3600	18	6
Industrial	West	Feb	20	880	5900	12	8
Residential	East	Feb	18	350	2300	25	5
Commercial	North	Mar	28	610	4100	20	7
Industrial	South	Mar	30	920	6200	15	9

Note: The R programs first create Energy_Consumption_Data.csv from the table above and then import it using read.csv().

PROGRAM 1: Histogram and Density Plot of Units Consumed

Import the dataset in R. Plot a histogram and density plot for Units_Consumed. Compare consumption patterns.

```
# Create the Energy_Consumption_Data.csv file
energy <- data.frame(
  Sector = c("Residential", "Commercial", "Industrial", "Residential", "Commercial",
"Industrial"),
  Region = c("North", "South", "West", "East", "North", "South"),
  Month = c("Jan", "Jan", "Feb", "Feb", "Mar", "Mar"),
  Temperature = c(15, 24, 20, 18, 28, 30),
  Units_Consumed = c(320, 540, 880, 350, 610, 920),
  Cost = c(2100, 3600, 5900, 2300, 4100, 6200),
  Renewable_Usage = c(22, 18, 12, 25, 20, 15),
  Peak_Hours = c(4, 6, 8, 5, 7, 9)
```

```

)
write.csv(energy, "Energy_Consumption_Data.csv", row.names = FALSE)

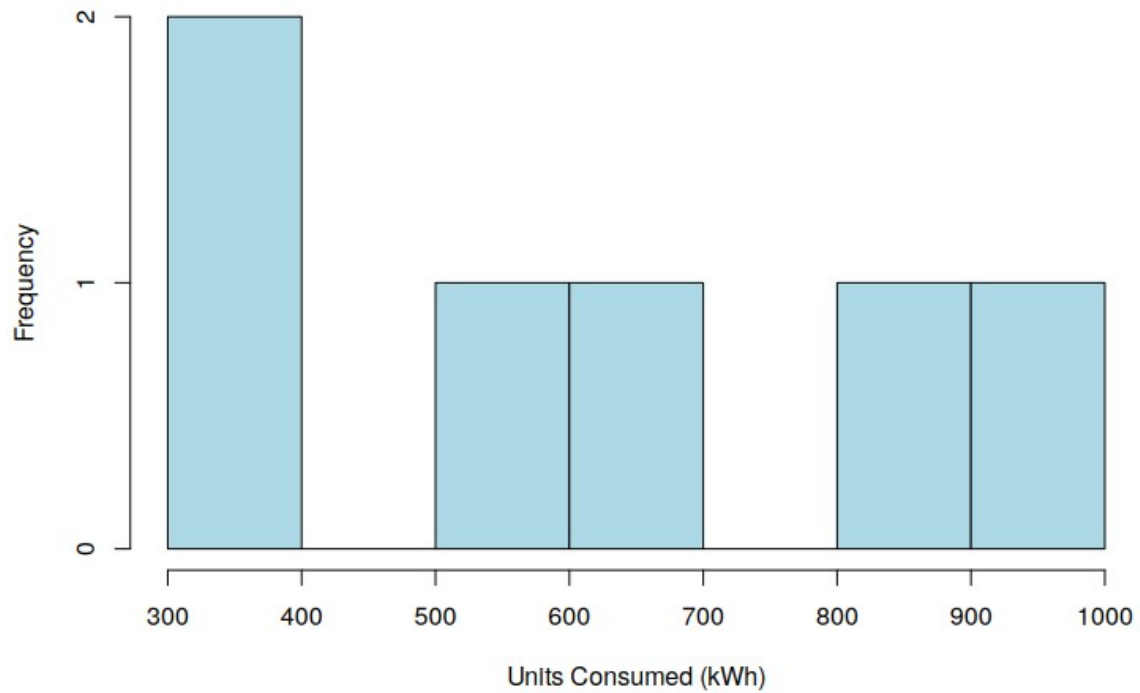
# Import the dataset
energy_data <- read.csv("Energy_Consumption_Data.csv")
print(energy_data)

# 1(a). Histogram of Units Consumed
hist(energy_data$Units_Consumed,
     breaks = seq(300, 1000, by = 100),
     col = "lightblue",
     border = "black",
     xlab = "Units Consumed (kWh)",
     ylab = "Frequency",
     main = "Histogram of Units Consumed")

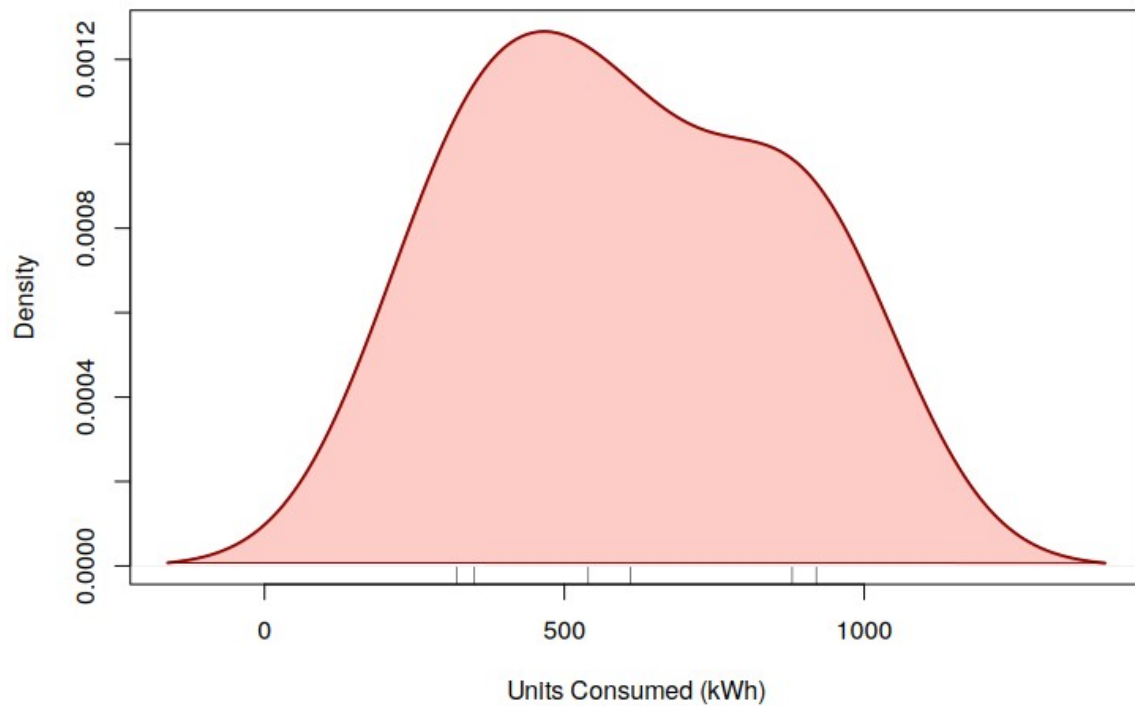
# 1(b). Density Plot of Units Consumed
dens <- density(energy_data$Units_Consumed)
plot(dens,
     col = "darkred",
     lwd = 2,
     xlab = "Units Consumed (kWh)",
     ylab = "Density",
     main = "Density Plot of Units Consumed")
polygon(dens, col = adjustcolor("salmon", alpha.f = 0.4), border = "darkred")
rug(energy_data$Units_Consumed, col = "black")

# Average consumption by sector
print(aggregate(Units_Consumed ~ Sector, data = energy_data, FUN = mean))

```

Histogram of Units Consumed

Output 1(a): Histogram of Units Consumed

Density Plot of Units Consumed

Output 1(b): Density Plot of Units Consumed

Comparison: The histogram shows three separate groups of consumption: 300–400 kWh (the two Residential records), 500–700 kWh (Commercial), and 800–1000 kWh (Industrial). The density plot shows the same pattern as a smooth curve with its main peak in the lower range and a second rise near 900 kWh. The average consumption is 603.3 kWh, but Industrial users consume about 900 kWh on average, which is about 2.7 times the Residential average of 335 kWh. Consumption depends strongly on the sector.

PROGRAM 2: Bubble Scatter Plot of Temperature vs Units Consumed

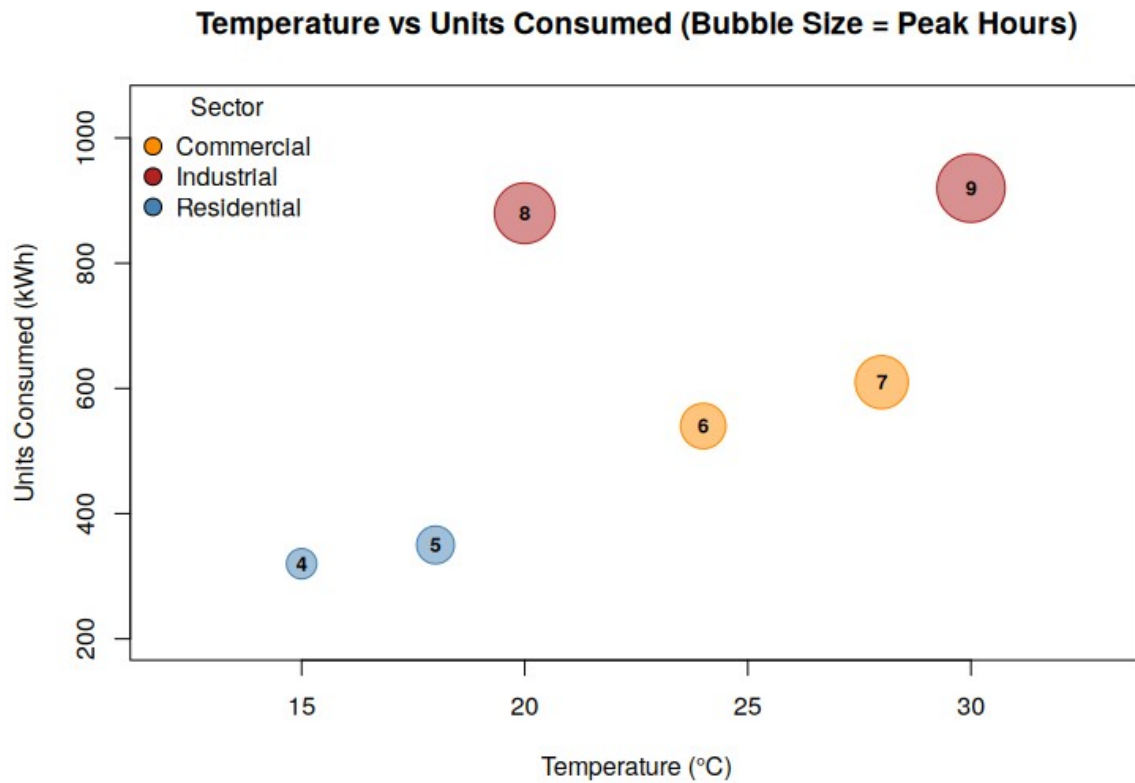
Using R programming, create a scatter plot between Temperature and Units_Consumed. Represent Peak_Hours using bubble size. Apply transparency.

```
# Create the Energy_Consumption_Data.csv file
energy <- data.frame(
  Sector = c("Residential", "Commercial", "Industrial", "Residential", "Commercial",
"Industrial"),
  Region = c("North", "South", "West", "East", "North", "South"),
  Month = c("Jan", "Jan", "Feb", "Feb", "Mar", "Mar"),
  Temperature = c(15, 24, 20, 18, 28, 30),
  Units_Consumed = c(320, 540, 880, 350, 610, 920),
  Cost = c(2100, 3600, 5900, 2300, 4100, 6200),
  Renewable_Usage = c(22, 18, 12, 25, 20, 15),
  Peak_Hours = c(4, 6, 8, 5, 7, 9)
)
write.csv(energy, "Energy_Consumption_Data.csv", row.names = FALSE)

# Import the dataset
energy_data <- read.csv("Energy_Consumption_Data.csv")
print(energy_data)

# 2. Bubble Scatter Plot: Temperature vs Units Consumed
sector_cols <- c(Commercial = "darkorange", Industrial = "firebrick", Residential =
"steelblue")
plot(energy_data$Temperature, energy_data$Units_Consumed,
  pch = 21,
  bg = adjustcolor(sector_cols[energy_data$Sector], alpha.f = 0.5), # transparency
  col = sector_cols[energy_data$Sector],
  cex = energy_data$Peak_Hours / 1.5, # bubble size
  xlim = c(12, 33), ylim = c(200, 1050),
  xlab = "Temperature (°C)",
  ylab = "Units Consumed (kWh)",
  main = "Temperature vs Units Consumed (Bubble Size = Peak Hours)")
text(energy_data$Temperature, energy_data$Units_Consumed,
  labels = energy_data$Peak_Hours, cex = 0.8, font = 2)
legend("topleft", legend = names(sector_cols), pt.bg = sector_cols,
  pch = 21, pt.cex = 1.5, bty = "n", title = "Sector")

# Correlation between temperature and units consumed
cor(energy_data$Temperature, energy_data$Units_Consumed)
```



Output 2: Temperature vs Units Consumed (Bubble Size = Peak Hours)

Observation: There is a moderate positive relationship between temperature and units consumed (correlation about 0.65). Larger bubbles, which mean more peak hours, appear at higher consumption levels. However, the Industrial record at 20°C still uses 880 kWh, so the sector affects consumption more than temperature does.

PROGRAM 3: Bar Chart of Average Renewable Usage by Sector

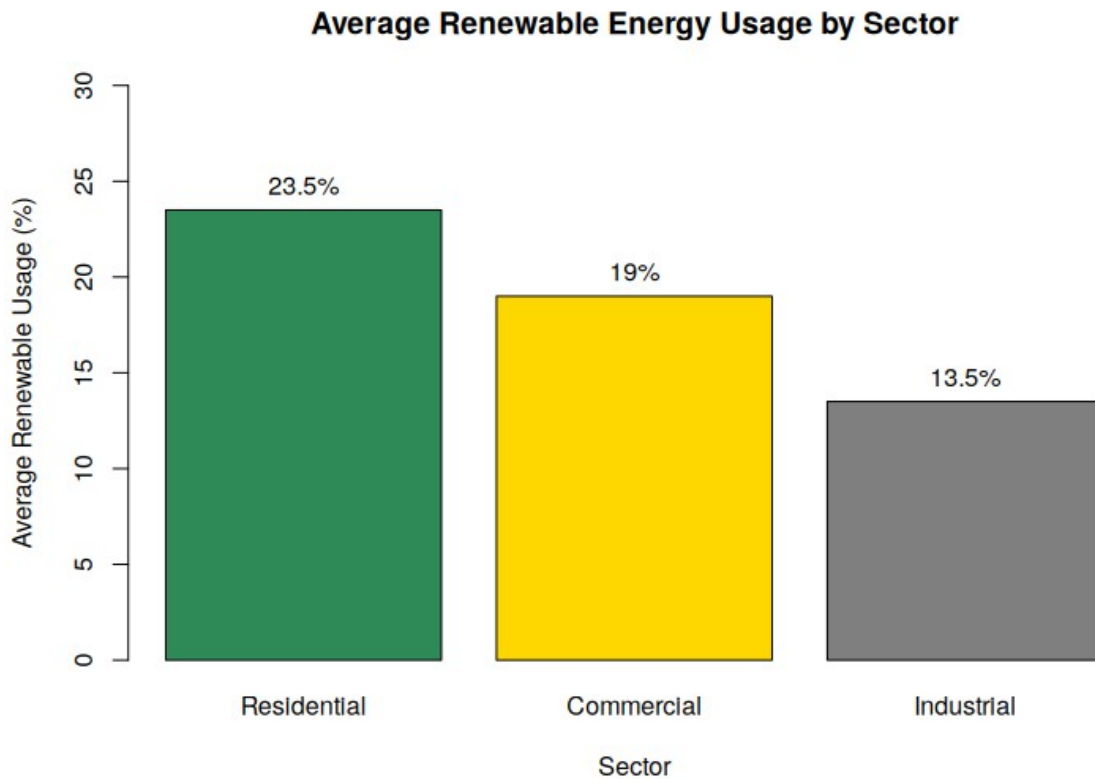
Using R, calculate average Renewable_Usage for each Sector. Create a bar chart. Interpret renewable energy adoption.

```
# Create the Energy_Consumption_Data.csv file
energy <- data.frame(
  Sector = c("Residential", "Commercial", "Industrial", "Residential", "Commercial",
"Industrial"),
  Region = c("North", "South", "West", "East", "North", "South"),
  Month = c("Jan", "Jan", "Feb", "Feb", "Mar", "Mar"),
  Temperature = c(15, 24, 20, 18, 28, 30),
  Units_Consumed = c(320, 540, 880, 350, 610, 920),
  Cost = c(2100, 3600, 5900, 2300, 4100, 6200),
  Renewable_Usage = c(22, 18, 12, 25, 20, 15),
  Peak_Hours = c(4, 6, 8, 5, 7, 9)
)
write.csv(energy, "Energy_Consumption_Data.csv", row.names = FALSE)

# Import the dataset
energy_data <- read.csv("Energy_Consumption_Data.csv")
print(energy_data)

# Average renewable usage for each sector
renew_avg <- aggregate(Renewable_Usage ~ Sector, data = energy_data, FUN = mean)
renew_avg <- renew_avg[order(-renew_avg$Renewable_Usage), ]
print(renew_avg)

# 3. Bar Chart of Average Renewable Usage by Sector
bp <- barplot(renew_avg$Renewable_Usage,
  names.arg = renew_avg$Sector,
  col = c("seagreen", "gold", "gray50"),
  xlab = "Sector",
  ylab = "Average Renewable Usage (%)",
  main = "Average Renewable Energy Usage by Sector",
  ylim = c(0, 30))
text(bp, renew_avg$Renewable_Usage, labels = paste0(renew_avg$Renewable_Usage, "%"), pos
= 3)
```



Output 3: Average Renewable Energy Usage by Sector

Interpretation: Residential users have the highest average renewable usage (23.5%), followed by Commercial (19%) and Industrial (13.5%). The sector that consumes the most energy (Industrial) has the lowest renewable adoption, so increasing renewable energy in industry would have the biggest impact.

PROGRAM 4: Interactive Dashboard for Energy Consumption

Import Energy_Consumption_Data.csv into Tableau. Create a line chart for monthly energy usage trends. Create a heatmap (Sector vs Region vs Units). Add filters for Month and Sector. Design an interactive dashboard.

Note: Tableau could not be installed on the system used for this experiment, so the interactive dashboard was developed in R using the Shiny package, with the same charts and interactive filters required in the question.

```
library(shiny)

# Dataset for Set 21 (Energy_Consumption_Data.csv)
energy_data <- data.frame(
  Sector = c("Residential", "Commercial", "Industrial", "Residential", "Commercial",
"Industrial"),
  Region = c("North", "South", "West", "East", "North", "South"),
  Month = c("Jan", "Jan", "Feb", "Feb", "Mar", "Mar"),
  Units_Consumed = c(320, 540, 880, 350, 610, 920)
)
months <- c("Jan", "Feb", "Mar")
sectors <- c("Commercial", "Industrial", "Residential")
regions <- c("East", "North", "South", "West")

ui <- fluidPage(
  titlePanel("Energy Consumption Dashboard"),
  sidebarLayout(
    sidebarPanel(width = 3,
      checkboxGroupInput("month", "Month:", choices = months, selected = months),
      checkboxGroupInput("sector", "Sector:", choices = sectors, selected = sectors)
    ),
    mainPanel(width = 9,
      fluidRow(
        column(6, plotOutput("linePlot", height = "340px")),
        column(6, plotOutput("heatPlot", height = "340px"))
      )
    )
  )
)

server <- function(input, output) {
  filtered <- reactive({
    subset(energy_data, Month %in% input$month & Sector %in% input$sector)
  })

  output$linePlot <- renderPlot({
    d <- filtered()
    total <- sapply(months, function(m) sum(d$Units_Consumed[d$Month == m]))
    plot(1:3, total, type = "o", col = "darkblue", lwd = 2, pch = 19, xaxt = "n", xlim =
c(0.7, 3.3),
      ylim = c(0, 1900), xlab = "Month", ylab = "Total Units Consumed (kWh)",
      main = "Monthly Energy Usage Trend")
    axis(1, at = 1:3, labels = months)
    text(1:3, total, labels = total, pos = 3)
    grid()
  }, width = 420, height = 340, res = 96)

  output$heatPlot <- renderPlot({
```

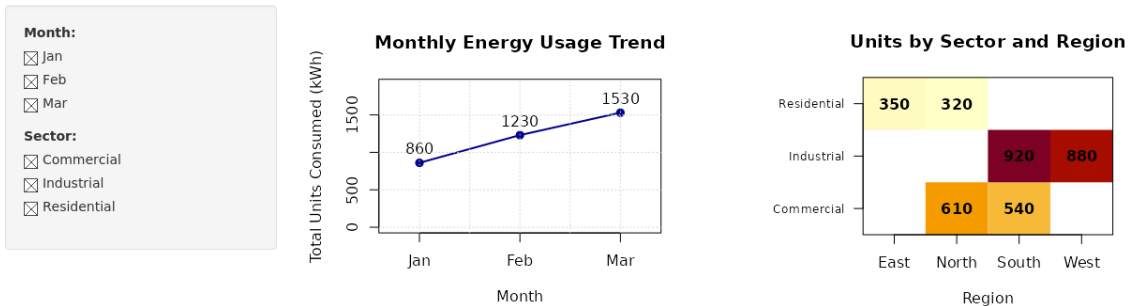
```

d <- filtered()
m <- matrix(NA, nrow = length(regions), ncol = length(sectors),
            dimnames = list(regions, sectors))
for (i in seq_len(nrow(d))) m[d$Region[i], d$Sector[i]] <- d$Units_Consumed[i]
par(mar = c(5, 6, 4, 2))
image(1:4, 1:3, m, col = hcl.colors(20, "YlOrRd", rev = TRUE), zlim = c(300, 950),
      axes = FALSE, xlab = "Region", ylab = "", main = "Units by Sector and Region")
axis(1, at = 1:4, labels = regions)
axis(2, at = 1:3, labels = sectors, las = 1, cex.axis = 0.8)
for (r in 1:4) for (s in 1:3)
  if (!is.na(m[r, s])) text(r, s, m[r, s], font = 2)
box()
}, width = 420, height = 340, res = 96)
}

shinyApp(ui = ui, server = server)

```

Energy Consumption Dashboard



Output 4: Energy Consumption Interactive Dashboard (R Shiny)

Observation: The dashboard shows the monthly energy usage trend and the sector–region heatmap side by side. Total usage rises from 860 kWh in January to 1230 kWh in February and 1530 kWh in March. The heatmap shows that Industrial users in the South and West regions consume the most energy. The Month and Sector filters update both views at the same time.

Procedure

1. Create the dataset, save it as `Energy_Consumption_Data.csv`, and import it using `read.csv()`.
2. Use `hist()` to plot the histogram and `density()` with `plot()` and `polygon()` to draw the density plot of `Units_Consumed`.
3. Use `plot()` with `cex` based on `Peak_Hours` and `adjustcolor()` for transparency to draw the bubble scatter plot, and `cor()` for the correlation.
4. Calculate the average `Renewable_Usage` per sector using `aggregate()` and draw the bar chart using `barplot()`.
5. Install and load the shiny package, and create the dashboard layout using `fluidPage()`, sidebar filters, and output areas for the charts and table.
6. Write the server function to filter the data with `reactive()` and redraw the outputs, then run the dashboard using `shinyApp()` and test the filters.

Result

The energy consumption data was successfully imported and visualized in R using a histogram, density plot, bubble scatter plot, and bar chart, and an interactive dashboard with a monthly trend line chart and a sector-region heatmap was designed using R Shiny.

Key R Functions Used

Function	Purpose
<code>write.csv()</code> / <code>read.csv()</code>	Creates and imports the CSV dataset.
<code>hist()</code>	Creates the histogram of units consumed.
<code>density()</code> / <code>polygon()</code>	Creates the filled density plot.
<code>adjustcolor()</code>	Applies transparency to the bubbles.
<code>cor()</code>	Calculates the correlation coefficient.
<code>aggregate()</code>	Calculates average values by sector.
<code>barplot()</code>	Creates the renewable usage bar chart.
<code>image()</code>	Draws the heatmap on the dashboard.
<code>fluidPage()</code> / <code>checkboxGroupInput()</code>	Creates the dashboard layout and filters.
<code>reactive()</code> / <code>renderPlot()</code>	Filters the data and redraws the charts.
<code>renderTable()</code>	Displays the filtered table on the dashboard.
<code>shinyApp()</code>	Runs the interactive dashboard.